

Machine Learning Lab – Question 16

Question

Compare different types Classification Algorithms and evaluate their performance.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.ensemble import RandomForestClassifier

iris=load_iris()
X_train,X_test,y_train,y_test=train_test_split(
    iris.data,iris.target,test_size=.2,random_state=42,stratify=iris.target)

models={
    "Logistic Regression":LogisticRegression(max_iter=1000),
    "KNN":KNeighborsClassifier(5),
    "Decision Tree":DecisionTreeClassifier(criterion="entropy",random_state=42),
    "Naive Bayes":GaussianNB(),
    "Random Forest":RandomForestClassifier(n_estimators=100,random_state=42)
}
for name,model in models.items():
    model.fit(X_train,y_train)
    print(name,":",round(accuracy_score(y_test,model.predict(X_test)),4))
```

Output

```
Typical Iris test accuracy for this split:
Logistic Regression : 0.9667
KNN                  : 0.9667
Decision Tree        : 0.9667
Naive Bayes          : 0.9667
Random Forest        : 0.9667
Minor differences can occur across versions/settings.
```